Phishing Email Detection & Awareness Report

Task 3 – Cyber Security

Phishing Email Detection & Awareness Report

Prepared By

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Track

Cyber Security (CS)

1. Executive Summary

This report analyzes a public API endpoint to identify security vulnerabilities. The analysis focuses on authentication issues, data exposure risks, and missing security controls.

2. Tools Used

- **Browser**
- **Public API testing**
- **Browser DevTools**
- **Documentation tools**

3. Target API

https://jsonplaceholder.typicode.com/users


```
[
  {
    "id": 1,
    "name": "Leanne Graham",
    "username": "Bret",
    "email": "Sincere@april.biz",
    "address": {
      "street": "Kulas Light",
      "suite": "Apt. 556",
      "city": "Gwenborough",
      "zipcode": "92998-3874",
      "geo": {
        "lat": "-37.3159",
        "lng": "81.1496"
      }
    },
    "phone": "1-770-736-8031 x56442",
    "website": "hildegard.org",
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    "company": {
      "name": "Romaguera-Crona",
      "catchPhrase": "Multi-layered client-server neural-net",
      "bs": "harness real-time e-markets"
    }
  },
  {
    "id": 2,
    "name": "Ervin Howell",
    "username": "Antonette",
    "email": "Shanna@melissa.tv",
    "address": {
      "street": "Victor Plains",
      "suite": "Suite 879",
      "city": "Wisokyburgh",
      "zipcode": "90566-7771",
      "geo": {
        "lat": "-43.9509",
        "lng": "-34.4618"
      }
    },
    "phone": "010-692-6593 x09125",
    "website": "anastasia.net",
    "company": {
      "name": "Deckow-Crist",
      "catchPhrase": "Proactive didactic contingency",
      "bs": "synergize scalable supply-chains"
    }
  },
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{
  "id": 3,
  "name": "Clementine Bauch",
  "username": "Samantha",
  "email": "Nathan@yesenia.net",
  "address": {
    "street": "Douglas Extension",
    "suite": "Suite 847",
    "city": "McKenziehaven",
    "zipcode": "59590-4157",
    "geo": {
      "lat": "-68.6102",
      "lng": "-47.0653"
    }
  },
  "phone": "1-463-123-4447",
  "website": "ramiro.info",
  "company": {
    "name": "Romaguera-Jacobson",
    "catchPhrase": "Face to face bifurcated interface",
    "bs": "e-enable strategic applications"
  }
},
{
  "id": 4,
  "name": "Patricia Lebsack",
  "username": "Karianne",
  "email": "Julianne.OConner@kory.org",
  "address": {

```
      "street": "Hoeger Mall",
      "suite": "Apt. 692",
      "city": "South Elvis",
      "zipcode": "53919-4257",
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        "lat": "29.4572",
        "lng": "-164.2990"
      }
    },
    "phone": "493-170-9623 x156",
    "website": "kale.biz",
    "company": {
      "name": "Robel-Corkery",
      "catchPhrase": "Multi-tiered zero tolerance productivity",
      "bs": "transition cutting-edge web services"
    }
  },
  {
    "id": 5,
    "name": "Chelsey Dietrich",
    "username": "Kamren",
    "email": "Lucio_Hettinger@annie.ca",
    "address": {
      "street": "Skiles Walks",
      "suite": "Suite 351",
      "city": "Roscoeview",
      "zipcode": "33263",
      "geo": {
        "lat": "-31.8129",
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```
      "lng": "62.5342"
    }
  },
  "phone": "(254)954-1289",
  "website": "demarco.info",
  "company": {
    "name": "Keebler LLC",
    "catchPhrase": "User-centric fault-tolerant solution",
    "bs": "revolutionize end-to-end systems"
  }
},
{
  "id": 6,
  "name": "Mrs. Dennis Schulist",
  "username": "Leopoldo_Corkery",
  "email": "Karley_Dach@jasper.info",
  "address": {
    "street": "Norberto Crossing",
    "suite": "Apt. 950",
    "city": "South Christy",
    "zipcode": "23505-1337",
    "geo": {
      "lat": "-71.4197",
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  },
  "phone": "1-477-935-8478 x6430",
  "website": "ola.org",
  "company": {
```

"name": "Considine-Lockman",
"catchPhrase": "Synchronised bottom-line interface",
"bs": "e-enable innovative applications"
}
},
{
"id": 7,
"name": "Kurtis Weissnat",
"username": "Elwyn.Skiles",
"email": "Telly.Hoeger@billy.biz",
"address": {
"street": "Rex Trail",
"suite": "Suite 280",
"city": "Howemouth",
"zipcode": "58804-1099",
"geo": {
"lat": "24.8918",
"lng": "21.8984"
}
},
"phone": "210.067.6132",
"website": "elvis.io",
"company": {
"name": "Johns Group",
"catchPhrase": "Configurable multimedia task-force",
"bs": "generate enterprise e-tailers"
}
},
{

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  "id": 8,
  "name": "Nicholas Runolfsdottir V",
  "username": "Maxime_Nienow",
  "email": "Sherwood@rosamond.me",
  "address": {
    "street": "Ellsworth Summit",
    "suite": "Suite 729",
    "city": "Aliyaview",
    "zipcode": "45169",
    "geo": {
      "lat": "-14.3990",
      "lng": "-120.7677"
    }
  },
  "phone": "586.493.6943 x140",
  "website": "jacynthe.com",
  "company": {
    "name": "Abernathy Group",
    "catchPhrase": "Implemented secondary concept",
    "bs": "e-enable extensible e-tailers"
  }
},
{
  "id": 9,
  "name": "Glenna Reichert",
  "username": "Delphine",
  "email": "Chaim_McDermott@dana.io",
  "address": {
    "street": "Dayna Park",
```

```
    "suite": "Suite 449",
    "city": "Bartholomebury",
    "zipcode": "76495-3109",
    "geo": {
      "lat": "24.6463",
      "lng": "-168.8889"
    }
  },
  "phone": "(775)976-6794 x41206",
  "website": "conrad.com",
  "company": {
    "name": "Yost and Sons",
    "catchPhrase": "Switchable contextually-based project",
    "bs": "aggregate real-time technologies"
  }
},
{
  "id": 10,
  "name": "Clementina DuBuque",
  "username": "Moriah.Stanton",
  "email": "Rey.Padberg@karina.biz",
  "address": {
    "street": "Kattie Turnpike",
    "suite": "Suite 198",
    "city": "Lebsackbury",
    "zipcode": "31428-2261",
    "geo": {
      "lat": "-38.2386",
      "lng": "57.2232"
```

```
      }
    },
    "phone": "024-648-3804",
    "website": "ambrose.net",
    "company": {
      "name": "Hoeger LLC",
      "catchPhrase": "Centralized empowering task-force",
      "bs": "target end-to-end models"
    }
  }
]
```


4. Identified Security Risks

1. Missing Authentication

The API endpoint allows access without authentication.

Risk: Unauthorized users can access sensitive data.

2. Data Exposure

The API response exposes user information such as:

- **Name**
- **Email**
- **Address**
- **Phone**

Risk: Possible privacy breach.

3. No Rate Limiting

The API does not restrict the number of requests.

Risk: Attackers may abuse the API or perform denial-of-service attacks.

5. Recommendations

- **Implement API authentication (API keys or tokens)**
- **Limit sensitive data exposure**
- **Enable rate limiting**
- **Add input validation**

6. Conclusion

API security is critical for protecting user data and maintaining system integrity. Implementing proper authentication and security controls can significantly reduce security risks.